

Riesz representation Theorem.

Thm. Let V be a fin. dim. inn. prod. v.s.

For any linear functional $f: V \rightarrow F$, there is a unique vector $\beta \in V$ s.t.

$$\text{for all } \alpha \in V, \quad f(\alpha) = \langle \alpha, \beta \rangle$$

In other words, $\varphi: V \rightarrow V^*: \beta \mapsto \langle \cdot, \beta \rangle$ is a Conjugate isomorphism.

Proof. Let $\{\alpha_1, \dots, \alpha_n\}$ be an orthonormal basis for V .

Given linear functional $f: V \rightarrow F$, put

$$\beta = \sum_{j=1}^n \overline{f(\alpha_j)} \cdot \alpha_j.$$

then, for each $1 \leq k \leq n$,

$$\langle \alpha_k, \beta \rangle = \left\langle \alpha_k, \sum_{j=1}^n \overline{f(\alpha_j)} \alpha_j \right\rangle = \sum_{j=1}^n \overline{f(\alpha_j)} \cdot \langle \alpha_k, \alpha_j \rangle = \overline{f(\alpha_k)} = f(\alpha_k)$$

Clearly, $\langle \cdot, \beta \rangle$ is linear, thus the existence is proved. And the uniqueness is shown:

if $\langle \alpha, \beta \rangle = \langle \alpha, \gamma \rangle$ for all $\alpha \in V$, then for $\alpha = \beta - \gamma$, $\langle \beta - \gamma, \beta - \gamma \rangle = 0$.

Thus, $\beta = \gamma$. $\square$

Observe that a inn. prod. space V with orthonormal basis $\{\alpha_1, \dots, \alpha_n\}$

Let $\alpha = x_1 \alpha_1 + \dots + x_n \alpha_n$ and $\beta = y_1 \alpha_1 + \dots + y_n \alpha_n$. Then,

$$\langle \alpha, \beta \rangle = x_1 \overline{y_1} + \dots + x_n \overline{y_n}$$

Meanwhile, for any linear functional $f \in V^*$,

$$f(\alpha) = f(\alpha_1)x_1 + \dots + f(\alpha_n)x_n$$

If we want to β satisfies $\langle \alpha, \beta \rangle = f(\alpha)$, then take $y_i = \overline{f(\alpha_i)}$. Then is,

$$\beta = \overline{f(\alpha_1)} \alpha_1 + \dots + \overline{f(\alpha_n)} \alpha_n$$

Geometric Observation.

Given a linear functional $f: V \rightarrow F$, $V = \ker f \oplus (\ker f)^\perp$.

Let P be the orthogonal projection of V on $(\ker f)^\perp$. Then,

for all $\alpha \in V$, $f(\alpha) = f(P\alpha)$.

Now, if $f \neq 0$, then $\text{rank } f = 1$, so for any non-zero vector $r \in (\ker f)^\perp$,

$$P\alpha = \frac{\langle \alpha, r \rangle}{\|r\|^2} \cdot r \text{ for all } \alpha \in V.$$

Finally, for given $\alpha \in V$,

$$f(\alpha) = f(P\alpha) = \langle \alpha, r \rangle \cdot \frac{f(r)}{\|r\|^2}, \text{ and } \beta = \frac{\overline{f(r)}}{\|r\|^2} \cdot r.$$